TechCorp Enterprise Network

Network Administration Portfolio Project

Network Adminstration

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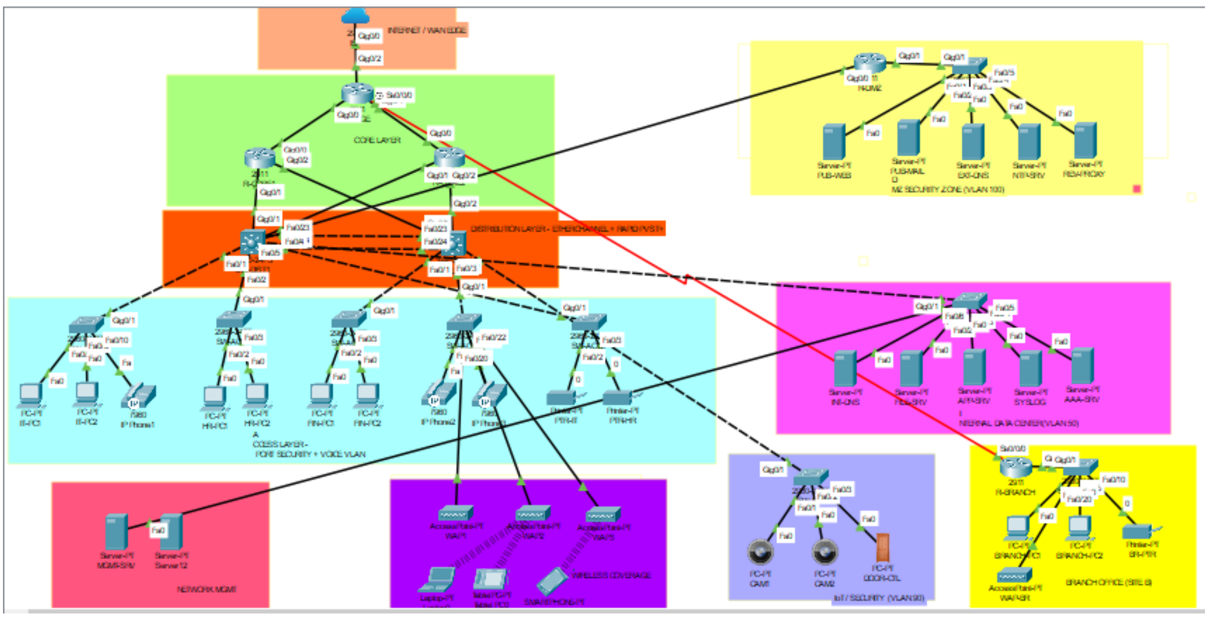


1. Project Overview

This project demonstrates a comprehensive enterprise network designed for TechCorp, a fictional technology company. The network implements industry-standard practices for security, redundancy, and scalability using Cisco Packet Tracer. The design encompasses a headquarters facility with multiple departments, a DMZ for public-facing servers, an IoT zone for security cameras, and a remote branch office connected via WAN.

The network serves approximately 50+ devices across 12 VLANs, providing logical segmentation for Management, IT, HR, Finance, Servers, Voice, Wireless (Corporate and Guest), IoT, DMZ, and Printers. Key features include HSRP redundancy for high availability, EtherChannel for increased bandwidth, Router-on-a-Stick for inter-VLAN routing, NAT/PAT for internet access, and comprehensive security through ACLs, port security, and SSH remote management.

Network Summary

Component	Details
Total Devices	51 devices (Routers, Switches, Servers, PCs, Printers, WAPs, IoT)
Routers	6 (R-EDGE, R-CORE1, R-CORE2, R-DMZ, R-BRANCH, ISP)
Switches	11 (2 Distribution, 5 Access, Server, DMZ, IoT, Branch)
VLANs	12 VLANs (10-Management, 20-IT, 30-HR, 40-Finance, 50-Servers, 60-Voice, 70-Wireless, 80-Guest, 90-IoT, 99-Native, 100-DMZ, 110-Printers)
IP Addressing	172.16.0.0/16 (HQ), 10.1.0.0/16 (Branch), 192.168.100.0/24 (DMZ), 203.0.113.0/30 (WAN)
Key Features	HSRP, EtherChannel, Router-on-a-Stick, NAT/PAT, ACLs, Port Security, SSH, DHCP

2. Network Design and Implementation

2.1 Network Topology

The network follows a hierarchical three-tier architecture consisting of Core, Distribution, and Access layers. This design provides scalability, redundancy, and clear separation of network functions. The topology includes an Edge router for WAN/Internet connectivity, dual Core routers with HSRP for redundancy, Distribution switches with EtherChannel for aggregation, and Access switches for end-device connectivity.

2.2 IP Addressing Scheme

The network uses a structured IP addressing scheme with clear subnet allocation for each department and function. The headquarters uses the 172.16.0.0/16 address space subdivided into /24 networks per VLAN. The branch office uses 10.1.0.0/16, and the DMZ uses 192.168.100.0/24 for public-facing servers.

VLAN	Name	Network	Gateway (HSRP VIP)
10	Management	172.16.10.0/24	172.16.10.1
20	IT Department	172.16.20.0/24	172.16.20.1
30	HR Department	172.16.30.0/24	172.16.30.1
40	Finance Department	172.16.40.0/24	172.16.40.1
50	Servers	172.16.50.0/24	172.16.50.1
100	DMZ	192.168.100.0/24	192.168.100.1

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:BCFF:FEC0:C08B
   IPv6 Address....................: ::
   IPv4 Address....................: 172.16.20.22
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     172.16.20.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

2.3 VLAN Configuration

VLANs provide logical segmentation of the network, improving security and reducing broadcast domains. Each department operates in its own VLAN, preventing unauthorized access between segments. The native VLAN (99) is configured as unused for security best practices, preventing VLAN hopping attacks.

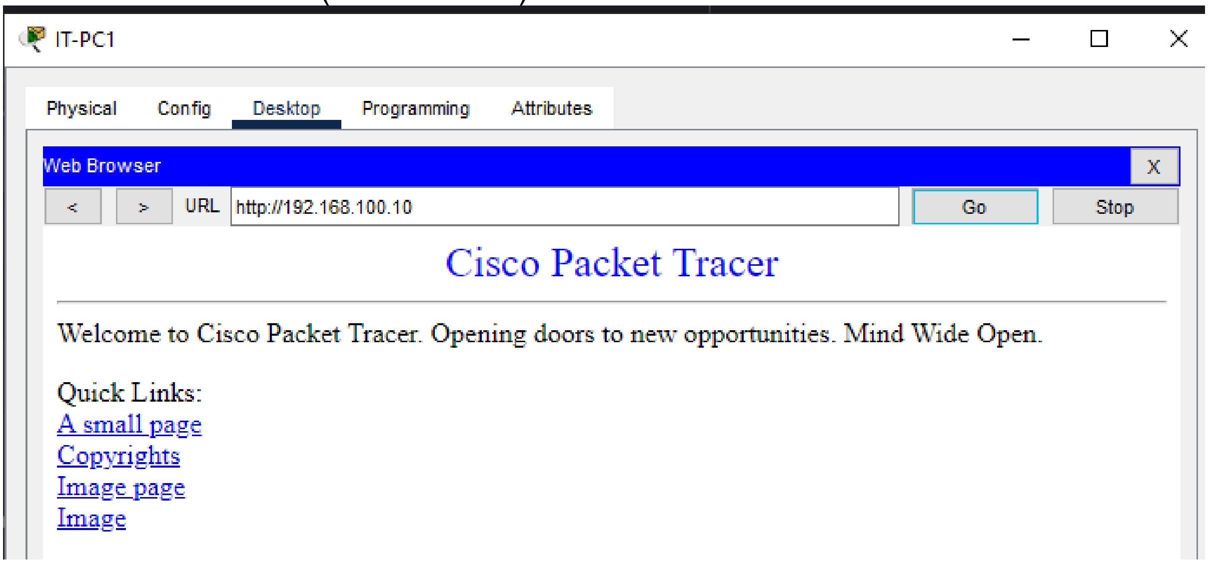
```
SW-DIST1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22
10   Management                       active
20   IT_Dept                          active
30   HR_Dept                          active
40   Finance_Dept                     active
50   Servers                          active
60   Voice                            active
70   Wireless_Corp                    active
80   Guest                            active
90   IoT                              active
99   Native_Unused                    active
100  DMZ                              active
101  DMZ_Transit                      active    Fa0/4
110  Printers                         active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

3. Protocol Configuration

3.1 Application Layer Services

The network implements several application layer services to demonstrate enterprise functionality:

1. **HTTP/HTTPS (Web Server):** PUB-WEB server (192.168.100.10) in the DMZ hosts the company's public website. Internal APP-SRV (172.16.50.30) hosts the intranet portal.
2. **SMTP/POP3 (Email Server):** PUB-MAIL server (192.168.100.20) provides email services with domain techcorp.com. Users can send and receive email from any workstation.
3. **FTP (File Transfer):** FILE-SRV (172.16.50.20) provides file storage with user authentication. IT users have full access while HR users have read-only permissions.
4. **DNS (Name Resolution):** EXT-DNS (192.168.100.30) handles external DNS, while INT-DNS (172.16.50.10) resolves internal hostnames.



3.2 Inter-VLAN Routing (Router-on-a-Stick)

Inter-VLAN routing is implemented using the Router-on-a-Stick method on R-CORE1 and R-CORE2. Each router has subinterfaces on GigabitEthernet0/1 configured with 802.1Q encapsulation for each VLAN. This allows traffic to flow between VLANs while maintaining logical separation. The HSRP virtual IP serves as the default gateway for all VLANs.

```
R-CORE1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.0.1.2        YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  up                    up
GigabitEthernet0/1.10  172.16.10.2     YES manual up                    up
GigabitEthernet0/1.20  172.16.20.2     YES manual up                    up
GigabitEthernet0/1.30  172.16.30.2     YES manual up                    up
GigabitEthernet0/1.40  172.16.40.2     YES manual up                    up
GigabitEthernet0/1.50  172.16.50.2     YES manual up                    up
GigabitEthernet0/1.60  172.16.60.2     YES manual up                    up
GigabitEthernet0/1.70  172.16.70.2     YES manual up                    up
GigabitEthernet0/1.80  172.16.80.2     YES manual up                    up
GigabitEthernet0/1.90  172.16.90.2     YES manual up                    up
GigabitEthernet0/1.101 10.0.100.1      YES manual up                    up
GigabitEthernet0/1.110 172.16.110.2    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/0/0   unassigned      YES unset  administratively down down
FastEthernet0/3/0      unassigned      YES unset  administratively down down
FastEthernet0/3/1      unassigned      YES unset  administratively down down
FastEthernet0/3/2      unassigned      YES unset  administratively down down
FastEthernet0/3/3      unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

3.3 Static Routing

Static routes are configured to establish connectivity between network segments. R-EDGE has a default route to the ISP (203.0.113.2) and static routes to internal networks via R-CORE1/R-CORE2. A static route to the DMZ network (192.168.100.0/24) directs traffic through R-DMZ for proper security zone enforcement.

```
R-EDGE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 7 subnets, 3 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/0
L       10.0.1.1/32 is directly connected, GigabitEthernet0/0
C       10.0.2.0/30 is directly connected, GigabitEthernet0/1
L       10.0.2.1/32 is directly connected, GigabitEthernet0/1
C       10.0.99.0/30 is directly connected, Serial0/0/0
L       10.0.99.1/32 is directly connected, Serial0/0/0
S       10.1.0.0/16 [1/0] via 10.0.99.2
S    172.16.0.0/16 [1/0] via 10.0.1.2
S    192.168.100.0/24 [1/0] via 10.0.1.2
     203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/30 is directly connected, GigabitEthernet0/2
L       203.0.113.1/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 203.0.113.2
```

4. Network Security Configuration

4.1 NAT and PAT Configuration

Network Address Translation is implemented on R-EDGE to allow internal devices to access the internet. PAT (Port Address Translation) overloads the outside interface (G0/2) with access-list 10 permitting internal networks. Static NAT entries map DMZ servers to public IP addresses for external access:

- PUB-WEB: 192.168.100.10 → 203.0.113.10
- PUB-MAIL: 192.168.100.20 → 203.0.113.20
- EXT-DNS: 192.168.100.30 → 203.0.113.30

```
R-EDGE#show ip nat translations
Pro  Inside global      Inside local       Outside local      Outside global
---  203.0.113.10       192.168.100.10     ---                ---
---  203.0.113.20       192.168.100.20     ---                ---
---  203.0.113.30       192.168.100.30     ---                ---
```

4.2 Access Control Lists (ACLs)

ACLs are implemented to control traffic flow and enforce security policies. Standard ACL 10 on R-EDGE defines networks permitted for NAT translation. Extended ACLs could be added on R-DMZ to restrict DMZ server access, allowing only HTTP/HTTPS to web servers and SMTP/POP3 to mail servers while blocking all other traffic from the internet.

```
R-EDGE#show access-lists
Standard IP access list 10
    10 permit 172.16.0.0 0.0.255.255
    20 permit 10.1.0.0 0.0.255.255
    30 permit 10.0.0.0 0.255.255.255
    40 permit 192.168.100.0 0.0.0.255
```

4.3 Port Security

Port security is configured on all access switch ports to prevent unauthorized device connections. Each port allows a maximum of 1-3 MAC addresses using sticky learning. Violation actions vary by department: Finance uses 'shutdown' for maximum security, while IT uses 'restrict' for flexibility. This prevents MAC flooding attacks and unauthorized device connections.

```
SW-ACC3#show port-security interface fa0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0060.3EAA.012C:40
Security Violation Count   : 0
```

4.4 SSH Configuration

All network devices are configured for secure remote management via SSH version 2. RSA keys with 2048-bit modulus are generated for encryption. Local authentication uses the 'admin' account with privilege level 15. VTY lines are configured to accept only SSH connections, with telnet disabled for security. Console ports require password authentication with logging synchronous enabled.

```
R-EDGE#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
```

5. Advanced Features

5.1 DHCP Configuration

DHCP is centrally configured on R-CORE1 to provide dynamic addressing across all user VLANs. Each VLAN has a dedicated DHCP pool with appropriate network, default-router (HSRP VIP), and DNS server settings. Address exclusions reserve the first 20 addresses in each subnet for static assignments (gateways, servers, printers). The Voice VLAN pool includes Option 150 for IP phone TFTP configuration.

```
R-CORE1#show ip dhcp binding
IP address        Client-ID/              Lease expiration        Type
                  Hardware address
172.16.20.22      00D0.BCC0.C08B           --                     Automatic
172.16.20.21      0090.2B85.9745           --                     Automatic
172.16.30.21      0090.2121.0BC4           --                     Automatic
172.16.30.22      000A.F386.6468           --                     Automatic
172.16.40.22      0002.4A90.5178           --                     Automatic
172.16.40.21      0060.3EAA.012C           --                     Automatic
172.16.60.21      0001.C935.91B7           --                     Automatic
172.16.70.21      0040.0B3C.27AA           --                     Automatic
172.16.70.23      0001.4342.DB86           --                     Automatic
172.16.80.22      0006.2AE7.B427           --                     Automatic
```

5.2 HSRP (Hot Standby Router Protocol)

HSRP provides gateway redundancy between R-CORE1 and R-CORE2. R-CORE1 is configured as the Active router with priority 110, while R-CORE2 serves as Standby with priority 100. Both routers share a virtual IP address (.1 in each VLAN) that serves as the default gateway. If R-CORE1 fails, R-CORE2 automatically assumes the Active role, ensuring continuous network access.

```
R-CORE1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   110 P Active   local           172.16.10.3     172.16.10.1
Gig         20   110 P Active   local           172.16.20.3     172.16.20.1
Gig         30   110 P Active   local           172.16.30.3     172.16.30.1
Gig         40   110 P Active   local           172.16.40.3     172.16.40.1
Gig         50   110 P Active   local           172.16.50.3     172.16.50.1
Gig         60   110 P Active   local           172.16.60.3     172.16.60.1
Gig         70   110 P Active   local           172.16.70.3     172.16.70.1
Gig         80   110 P Active   local           172.16.80.3     172.16.80.1
Gig         90   110 P Active   local           172.16.90.3     172.16.90.1
Gig         110  110 P Active   local           172.16.110.3    172.16.110.1
```

5.3 EtherChannel

EtherChannel is configured between SW-DIST1 and SW-DIST2 using LACP (Link Aggregation Control Protocol). Ports FastEthernet0/23-24 on both switches are bundled into Port-channel 1, providing 200 Mbps aggregate bandwidth and link redundancy. If one link fails, traffic continues over the remaining link without interruption.

```
SW-DIST1#show etherchannel summary
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+----------------------------------------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
```

5.4 Spanning Tree Protocol

Rapid PVST+ is configured across all switches for loop prevention. SW-DIST1 is designated as the root bridge (root primary) for all VLANs, while SW-DIST2 serves as the secondary root. Access switch ports connected to end devices have PortFast enabled for immediate forwarding, and BPDUGuard is enabled on Finance department ports for enhanced security.

```
SW-DIST1#show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: Management IT_Dept HR_Dept Finance_Dept Servers Voice Wireless_Corp
Guest IoT Native_Unused DMZ DMZ_Transit Printers
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is disabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                     Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                     3         0        0          7         10
VLAN0020                     6         0        0          4         10
VLAN0030                     6         0        0          4         10
VLAN0040                     7         0        0          3         10
VLAN0050                     6         0        0          4         10
VLAN0060                     6         0        0          4         10
VLAN0070                     7         0        0          3         10
VLAN0080                     7         0        0          3         10
VLAN0090                     7         0        0          3         10
VLAN0099                     1         0        0          9         10
VLAN0100                     7         0        0          3         10
VLAN0101                     4         0        0          6         10
VLAN0110                     4         0        0          6         10

---------------------- -------- --------- -------- ---------- ----------
14 vlans                    71         0        0         59        130
```

6. Testing and Verification

6.1 Connectivity Tests

Comprehensive testing verified network functionality across all segments:

Test	Command/Action	Result
DHCP Address Assignment	ipconfig on IT-PC1	**PASS**
Gateway Reachability	ping 172.16.20.1	**PASS**
Inter-VLAN Routing	ping 172.16.50.10	**PASS**
Internet Connectivity (NAT)	ping 8.8.8.8	**PASS**
Web Server Access	Browser: 192.168.100.10	**PASS**
Branch WAN Connectivity	ping 172.16.50.10 from BR-PC1	**PASS**

7. Challenges and Lessons Learned

7.1 Challenges Faced

Router Module Limitations

The Cisco 2911 routers in Packet Tracer have limited built-in interfaces. When additional ports were needed for the DMZ and server connections, HWIC-4ESW modules were added. However, these modules provide switch ports rather than routable interfaces, requiring architecture redesign to route DMZ traffic through the distribution layer instead of directly through Core routers.

HSRP Communication Issues

Initially, both R-CORE1 and R-CORE2 showed 'Active' status with 'unknown' standby partners. This indicated the routers couldn't communicate HSRP hello packets. The solution required ensuring proper trunk configuration on distribution switches with VLAN 101 added to allowed VLANs, enabling HSRP traffic to flow between the routers.

DMZ Routing Complexity

Establishing connectivity to the DMZ proved challenging due to overlapping subnet issues and interface assignment problems. The R-DMZ router required careful configuration with separate inside (10.0.100.0/24) and outside (192.168.100.0/24) interfaces. A dedicated VLAN 101 transit network was created to properly route traffic between the internal network and DMZ zone.

EtherChannel Configuration

The EtherChannel between distribution switches initially showed 'SD' (Suspended/Down) status with individual ports. Adding 'switchport trunk encapsulation dot1q' to both physical interfaces and the Port-channel resolved the issue, bringing the bundle to 'SU' (Layer 2, In Use) status.

7.2 Lessons Learned

1. **Plan Before Implementation:** Thoroughly planning the network architecture, including all required interfaces and VLANs, before beginning configuration prevents major redesigns mid-project.
2. **Verify Layer by Layer:** Testing connectivity at each layer (physical, data link, network) helps isolate issues quickly. Using 'show' commands extensively aids troubleshooting.
3. **Trunk Configuration Matters:** Proper trunk configuration with correct encapsulation and allowed VLANs is critical for VLAN traffic and redundancy protocols like HSRP.
4. **Documentation is Essential:** Maintaining clear documentation of IP addressing, VLAN assignments, and port connections significantly reduces troubleshooting time.

7.3 Alternative Approaches Considered

1. **Layer 3 Switches:** Using Layer 3 switches for inter-VLAN routing instead of Router-on-a-Stick would provide better performance but wasn't chosen to demonstrate router configuration skills.

2. **Dynamic Routing:** OSPF or EIGRP could replace static routes for automatic path selection, but static routing was used per project requirements.
3. **Simplified DMZ:** Connecting SW-DMZ directly to distribution switches without R-DMZ was considered but rejected to maintain proper security zone architecture.

8. Conclusion

This project successfully demonstrates a comprehensive enterprise network implementation incorporating essential networking concepts. The TechCorp network provides departmental segmentation through VLANs, high availability via HSRP redundancy, secure remote management through SSH, internet connectivity with NAT/PAT, and proper security zone architecture with the DMZ. The challenges encountered during implementation provided valuable learning experiences in troubleshooting and network design. This project reinforces the importance of systematic planning, thorough testing, and comprehensive documentation in network administration.

Appendix: Address Calculation Reference

Subnet Calculations

Base Network: 172.16.0.0/16

Subnet Mask: 255.255.255.0 (/24)

Hosts per Subnet: 254 usable (2^8 - 2)

VLAN Subnets: 172.16.[VLAN_ID].0/24 (e.g., VLAN 20 = 172.16.20.0/24)

DHCP Range: .21-.254 (first 20 reserved for static assignments)